



Stockholm University

DIFO 2023 Lab-1 Answers

Group-11

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1.1 File type classification using a tool

Q1: How many photos/images are there in the dataset?

Answer: 288

Q2: What does the file -command say when it does not recognize a binary file format?

Answer: unknown_file: Data

Q3: How many files did the file -command not recognize?

Answer: 898

Q4: Write a command line that answers the above question (i.e. a line of text that can be pasted into the kali terminal) .

Answer: `file *.bin | grep -x -c '.*: *data'`

Q5: In at most 200 characters, explain briefly how that command line works.

Answer: `file *.bin`: It uses the file command to analyze and display information about all files in the current directory with a .bin extension. Then I use the grep command to search for lines that match the pattern '.*: *data' in the output from the file command - x in order to match the entire line and -c to count the number of matches.

Q6: Compare the first few bytes of the photo/image files you found. What are the three most common “magic numbers” you see?

Answer: 8950, 3c73, 3c3f

1.3 Text search

Q7: Which files contain the text “heffaklump” in either ASCII or UTF-8?

Answer: 2903.bin 2384.bin

Q8: Which files contain the text “heffaklump” in little endian UTF-16 encoding?

Answer: 3615.bin

Q9: Write a grep pattern to find IPV4-addresses (e.g. 193.123.156.189). (It does not have to be perfect, some false positives are okay.)

Answer: `grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b'`

Q10: How many unique IPv4-addresses can you find in the dataset? Use the pattern you described above.

Answer: total 2428

1.4 Hashing

Q11: Which groups of files in the dataset share the same SHA1 hash? (Tip: sort | uniq -c and grep)

Answer: The following command was used to identify groups of files in the dataset that share the same SHA1 hash:

```
find . -type f -name "*.bin" -exec sha1sum {} + | sort | uniq -d -w 40
```

Here's the corresponding output:

```
00e5f837ad5aa8a26394e8daf0c2da80cff359ec ./1448.bin
02d61a186293f7b5fc2c66b5dc3359ff6ebc683b ./1073.bin
2995201601ab916bd24c4700749965e1bfa82ec5 ./2535.bin
2fe398f1ebced166087362626241b95efeaab407 ./3585.bin
51f223779806aa6d23348a3aaf55b439c4592520 ./1083.bin
5bbb95a6ae5db17da14c72a95b79cb40eb5ecd5b ./3077.bin
8c85c0cda68488de9872ae5538b6998fb5da8ee5 ./424.bin
da39a3ee5e6b4b0d3255bfef95601890afd80709 ./1023.bin
```

Q12: What command line did you use to answer the above question?

Answer: `sha1sum *.bin | cut -d ' ' -f 1 | sort | uniq -c | sort -n`

Q13: Explain in at most 200 characters how that command line works.

Answer: "sha1sum *.bin:" This part of the command calculates the SHA-1 checksum for each file in the current directory that has a .bin extension. cut -d ' ' -f 1 the command extracts the first field from the output of sha1sum and sorts the extracted SHA-1 checksums in ascending order, making it easier to identify duplicates. Then it selects unique sha1sum checksums and counts them. Lastly, again sort results in ascending numerical order based on the counts.

Q14: There is something odd about the hashes of four files in the dataset. Which ones? (hint: compare sha1 and md5 using the command line you developed in the above questions).

Answer: 42.bin, 1559.bin, 558.bin, 3316.bin

Q15: Which files have the wrong hash, according to 3614.bin ?

Answer: 2384.bin

1.5 Visualize the difference between two files

Q16: Compare files 1315.bin and 2091.bin (using e.g. vimdiff). How many bytes differ?

Answer: 5 bytes

Q17: Write a python script to determine which files are in the database. Below is a stub you can start from (894.bin). Beware of upper and lower case hashes when comparing! It is probably fastest to use SHA256.

```
#!/usr/bin/env python3
import hashlib
import sqlite3
import pathlib
import argparse

def sha256_hash_file(path):
    sha256_hash = hashlib.sha256()
    with open(path, 'rb') as file:
        while True:
            data = file.read(65536)
            if not data:
                break
            sha256_hash.update(data)
    return sha256_hash.hexdigest()

def main():
    parser = argparse.ArgumentParser()
    parser.add_argument('database_file')
    parser.add_argument('files_to_hash', nargs='+')
    args = parser.parse_args()

    # Open the NSRL database
    conn = sqlite3.connect(args.database_file)
    cursor = conn.cursor()

    for path in args.files_to_hash:
        hash = sha256_hash_file(path)

        # Check if the hash is in the database
        cursor.execute("SELECT sha256 FROM DISTINCT_HASH WHERE sha256=?", (hash,))
        result = cursor.fetchone()

        if result is not None:
            print(path, hash)

    conn.close()

if __name__ == '__main__':
    main()
```

Then we run the script:

```
(cs2lab@eve12) - [~/difo]
```

```
$ python3 script.py ~/nsrl//RDS_2023.03.1_modern_minimal.db trash/*.bin
```

And

```
$ python3 script.py ~/nsrl//RDS_2023.03.1_modern_minimal.db trash/*.bin | wc -l
2059
```

```
$ file trash/*.bin | wc -l
3616
```

```
$ python3
>>>3616 - 2059
1557
>>>
```

Q18: How many files in the lab dataset are “uninteresting” according to the 2023.03.1 Modern PC minimal version of NSRL RDS?

Answer: Uninteresting files are 1557

Q19: What are the 10 most similar pairs of files in the dataset, excluding pairs that are exactly identical?

Answer:

```
cli: ssdeep -p *.bin | sort -k 4
```

Output

```
/home/samir/Desktop/labwork/trash/1559.bin matches /home/samir/Desktop/labwork/trash/42.bin (97)
/home/samir/Desktop/labwork/trash/254.bin matches /home/samir/Desktop/labwork/trash/2679.bin (97)
/home/samir/Desktop/labwork/trash/254.bin matches /home/samir/Desktop/labwork/trash/3090.bin (97)
/home/samir/Desktop/labwork/trash/2679.bin matches /home/samir/Desktop/labwork/trash/254.bin (97)
/home/samir/Desktop/labwork/trash/2679.bin matches /home/samir/Desktop/labwork/trash/3090.bin
(97)
/home/samir/Desktop/labwork/trash/2681.bin matches /home/samir/Desktop/labwork/trash/3429.bin
(97)
/home/samir/Desktop/labwork/trash/3090.bin matches /home/samir/Desktop/labwork/trash/254.bin (97)
/home/samir/Desktop/labwork/trash/3090.bin matches /home/samir/Desktop/labwork/trash/2679.bin
(97)
/home/samir/Desktop/labwork/trash/3429.bin matches /home/samir/Desktop/labwork/trash/2681.bin
(97)
/home/samir/Desktop/labwork/trash/42.bin matches /home/samir/Desktop/labwork/trash/1559.bin (97)
```

1.8 Photo Metadata extraction using Exiftool

Q20: Which photo files in the dataset were taken by a Samsung S9 smartphone camera (according to the EXIF metadata)?

Answer: 3 files:

1472.bin

1930.bin

2428.bin

Q21: Which files in the dataset contain GPS coordinates in their EXIF data?

Answer:

4 files.

Q22: For each of those files, which was the closest Swedish city where the photo taken?

Answer:

3 files:

1472.bin kista

1930.bin Norrtälje

2428.bin kista

1.9 Entropy

Q23: Which ZIP-files and which PDF-files are encrypted? Try opening each file one by one.

Answer: zip - 291.bin

pass - a.odt

PDF = evince 496.bin

evince 2673.bin

Q24: File 1924.bin is a unix-style “passwd” file and 651.bin is a unix-style “shadow” file (both from the same computer). What is the password of user drsnuggles ?

Answer: drsnuggles:password1

Q25: Files 2834.bin and 1596.bin consist of the “SYSTEM” and “SAM” registry files of a Windows 10 machine. What is the password of user inspectorgadget?

Answer: nelson

Q26: There is a password-protected PDF in the dataset. What is the title of that PDF (i.e. the title text written inside that PDF)?

Answer: aaab

Q27: There is a password-protected ZIP-file in the dataset. What is the total of John Doe?

Answer: 95

Q28: I have used steghide to hide a string of text in one of the JPEG files in the dataset. Which file?

Answer: stegseek -sf 693.bin
 "Evil_plan.txt"

Q29: What is the "fourth step" mentioned in the embedded file?

Answer: Move to thailand
 P - Cat 693.bin.out

2.1 Preservation

Q30: What is the sha1 hash of the disk?

Answer: 5172e16f810b3c571141eb162e4d3d361ea3fe62

Q31: What is the md5 hash of the disk?

Answer: 0bce88cb246525230e0d847acf1b1873

2.2 Viewing the Partition tables using mmls

Q32: What is the sector-offset of the largest partition?

Answer: 0000239616

Q33: What is the byte offset of the largest partition?

Answer: 122683392

Q34: How many sectors long is the largest partition?

Answer: 103543269 sectors

Q35: How many bytes long is the largest partition?

Answer: 53014153728

2.3 Viewing files using sleuthkit cli tools

Q36: How many (allocated) files are in the largest partition of the disk?

Answer: 313582

Q37: What is the contents of osver.txt ?

Answer: 10.0.19045

2.4 Carving MFT entries

Q38: Write a scalpel configuration file for carving only MFT entries.

Answer: MFT y 1024 FILE

Q39: How many MFT entries did you manage to carve?

Answer: Total:377733

Q40: Look at the MFT that was carved from byte-offset 3320834624 (it is easier to look at the file you extracted, but you can also look directly in the disk file). What is the modification timestamp of this MFT10?

Answer: 2023-09-21

Q41: Look at the MFT entry that was carved from byte-offset 3320834624. What is the filename of this MFT? Note that this MFT is a file, not a directory.

Answer: 00012510.MFT

Q42: Look at the MFT entry that was carved from byte-offset 3320834624. Is this a file or a directory?

Answer: file

Q43: How many JPEG-files did you find?

Answer: 770 jpg files

Q44: In the folder C:\difo_resident on the disk, there are four files. Which of these use resident \$DATA attributes?

Answer: a.txt and b.txt

Q45: In folder C:\difo_gone_small I have created a lot of files (1.txt, 2.txt, etc.) and then deleted some of them. What is the highest numbered deleted filename you can obtain from the directory's \$I30 metadata (aka. "index file"). Hint: use istat and icat .

Answer: 9

2.8 Windows RAM artifacts using Volatility

Q46: How many processes were running?

Answer: 91

Q47: What was the name of exe-file of process 1612?

Answer: curl.exe

Q48: There was a connection from ?.?.?.?:X to 193.10.9.5:443. What was port X?

Answer: Port: 49728

Q49: How many occurrences of the password dsvcs2 can you find in the blob? Hint: String

Answer: 65

Q50: There is a web browser installed (that is not made by Microsoft). What is its name?

Answer: Google Chrome v.117.0.5938.89

Q51: There is a deleted WEBP file in the recycle bin, what was its original name?

Answer: Himalayan-chocolate-point.jpg

Q52: There are two files in the C:\difo_carve_me -directory. According to their respective Zone-identifier-files, Where were the files downloaded from?

Answer: Wikimedia.org and google.com

Q53: How many hours is the timezone off from UTC?

Answer: 8 hours

Q54: List the search queries, one per line?

Answer: 2 queries
0- heffaklump
1-explorer searchhe ffaklump

Q55: When was the shell bag for C:\difo_gone_small last modified in local time?

Answer: 2023-09-21 06:58:48 CEST 3

Q56: When was the last time this system booted up?

Answer: 2023-09-21 09:17.58 CEST

Q57: Which normal user was the last to log in?

Answer: Target User name Cs2lab 2023-09-21 09:16:17

Q58: How many times has chrome.exe run, according to windows prefetch?

Answer: Total count: 62

4.3 Browser artifacts:

Google chrome Do a forensic experiment with google chrome. Start chrome in the windows VM, google for something unique (easy to search), for example "winniethepooh". Close google chrome. Now perform either a logical or physical acquisition. The former is probably faster and easier.

Q59: Search through your acquisition. What files seem to contain traces of your search?

Answer: User Data/Default/History
 User Data/Default/Shortcuts
 Grep -r -a -l "Cat"

Chrome/User Data/Default/Cache/Cache_Data/data_1
Chrome/User Data/Default/Favicons
Chrome/User Data/Default/History
Chrome/User Data/Default/Network Action Predictor
Chrome/User Data/Default/Sessions/Session_13342027629616160
Chrome/User Data/Default/Sessions/Session_13342030085930667
Chrome/User Data/Default/Sessions/Tabs_13342017603157535
Chrome/User Data/Default/Shortcuts
Chrome/User Data/Default/Sync Data/LevelDB/000079.log
Chrome/User Data/Default/WebStorage/11/IndexedDB/indexeddb.leveldb/000004.log